THE LAST FACE MAP PRESERVES THE NORMALIZATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the substitution: x in N(U_m) means d^m_j x = 0 for j < m, so d^{m-1}_j (d^m_m x) = d^{m-1}_{m-1} (d^m_j x) = 0 for all j <= m-2.

Editorial note. The label lemma-N-d-in-N names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Normalization splitting. Same unit, Lemma lemma-splitting-abelian-category. N(U_m) denotes the normalization of Lemma lemma-splitting-abelian-category.
- Simplicial identities. Same unit, Remark remark-relations. The identity d^{m-1}_j d^m_m = d^{m-1}_{m-1} d^m_j is one of the relations of Remark remark-relations.

Source reference. Original reference 017V: sources/stacks-simplicial/pinned-original-source.tex, source lines 2245–2245. Prerequisite pointer invoked at proof line(s) 2298; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 016C: sources/stacks-simplicial/pinned-original-source.tex, source lines 471–471. Prerequisite pointer invoked at proof line(s) 2305; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let U be a simplicial object in $\mathcal{A}$. Let $N(U_m)$ as in Lemma 017V above. Then $d_m^m(N(U_m)) \subset N(U_{m-1})$.*

Proof. For $j = 0, \ldots, m-2$ we have $d_j^{m-1} d_m^m = d_{m-1}^{m-1} d_j^m$ by the relations in Remark 016C. The result follows. □

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